

- **Eye diagram**

Mat lab code

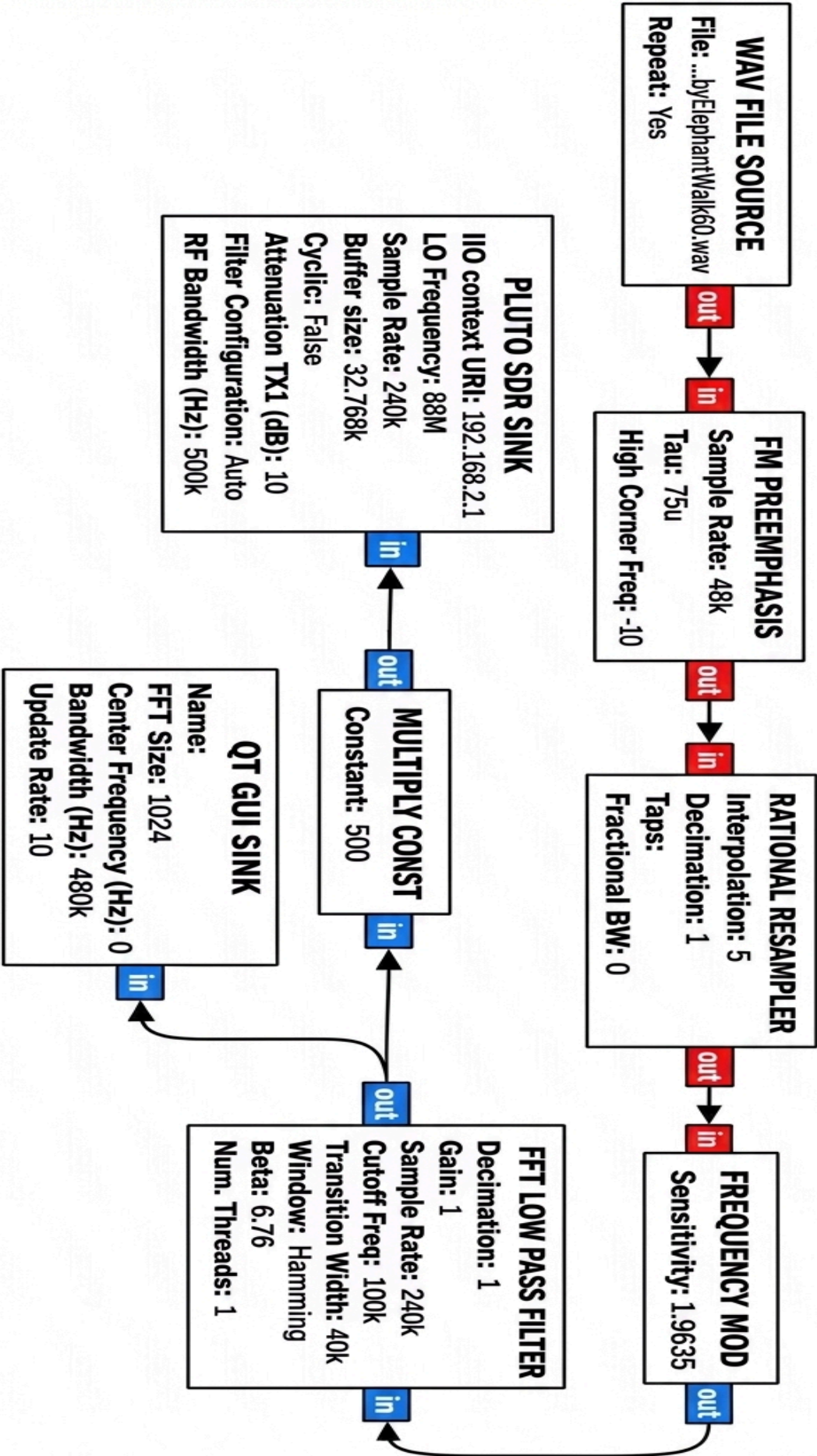
```
clc;
clear all;
close all;
rolloff_1=0.1;
rolloff_5=0.5;
span=10;
sps=8;
N=10;
BP_data=2*randi([0 1],N,1)-1;
rctfilt1=rcosdesign(rolloff_1,span,sps,'sqrt');
rctfilt5=rcosdesign(rolloff_5,span,sps,'sqrt');
UP_s1=upfirdn(BP_data,rctfilt1,sps,1);
UP_s5=upfirdn(BP_data,rctfilt5,sps,1);
figure;
subplot(2,1,1);
plot(UP_s1);
subplot(2,1,2);
plot(UP_s5);
figure;
eyediagram(UP_s1,2*sps);
title('alpha=0.1');
figure;
eyediagram(UP_s5,2*sps);
title('alpha=0.5');
```

OPTIONS

Title: FM Modulation
Author: STARDOM
Output Language: Python
Generate Options: QT GUI

VARIABLE

ID: samp_rate
Value: 48k



Options
Title: FM Modulation
Author: Sultan Mohyuddin
Output Language: Python
Generate Options: QT GUI

Variable
ID: samp_rate
Value: 480k

Variable
ID: freq
Value: 104.8M

PlutoSDR Source
IIO context URI: 192.168.2.1
LO Frequency: 101.1M
Sample Rate: 480k
Buffer size: 16.384k
Quadrature: True
RF DC Correction: True
BB DC Correction: True
Gain Mode (RX1): Slow Attack
Filter Configuration: Auto
RF Bandwidth (Hz): 500k

Low Pass Filter
Decimation: 2
Gain: 1
Sample Rate: 480k
Cutoff Freq: 100k
Transition Width: 44.1k
Window: Hamming

FM Demod
Channel Rate: 240k
Audio Decimation: 5
Deviation: 75k
Audio Pass: 15k
Audio Stop: 20k
Gain: 1
Tau: 75u

Audio Sink
Sample Rate: 48 kHz

QT GUI Range
ID: tun_freq
Label: UHD Freq (MHz)
Default Value: 104.8M
Start: 87.5M
Stop: 108M
Step: 100k

QT GUI Frequency Sink
FFT Size: 512
Center Frequency (Hz): ...freq
Bandwidth (Hz): 400k